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MENTAL HEALTH DETECTION USING MACHINE LEARNING

INTERNSHIP WORK SUBMITTED IN PARTIAL FULFILLMENT OF THE
REQUIREMENT FOR THE AWARD OF THE DEGREE

OF

BACHELOR OF COMPUTER SCIENCE WITH CYBER SECURITY

SUBMITTED

BY

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JUNE 2026

CERTIFICATE

This is to certify that the internship work entitled “**MENTAL HEALTH DETECTION USING MACHINE LEARNING**” submitted to **PSGR Krishnammal College for Women, Coimbatore**, in partial fulfillment of the requirement for the award of the degree of Bachelor of Computer Science with Cyber Security, is a record of the original work done by **Ms. Subhasri A (24SBCY056)** during her period of study in the Department of Computer Science with Cyber Security, PSGR Krishnammal College for Women, Coimbatore, under my supervision and guidance. Her internship work has not formed the basis for the award of any Degree / Diploma / Associate / Fellowship or any similar title to any candidate of any University.

Forwarded by

Mrs.SUDHANDRA DEVI P.,,M.Phil. Ph.D.,

Faculty guide

Mr.T.PRADEEP.,MCA.,Ph.D

Head of the Department

CERTIFICATE

WORK DIARY

DECLARATION

I hereby declare that the internship work entitled “**MENTAL HEALTH DETECTION USING MACHINE LEARNING**” submitted to Bharathiar University, Coimbatore, for the award of the Degree of Bachelor of Computer Science with Cyber Security, is a record of original work done by **Ms. Subhasri A (24SBCY056)** under the supervision and guidance of **Mrs.Sudhandra devi .P**, Department of Computer Science with Cyber Security, **PSGR Krishnammal College for Women, Coimbatore**, and that this internship work has not formed the basis for the award of any **Degree / Diploma / Associate Fellowship** or any similar title to any candidate of any University.

SUBHASRI A
(24SBCY056)

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ABSTRACT

The framework integrates an interactive web user interface, machine learning model serialization, and automated localized report generation to identify individuals requiring preliminary support or professional consultation. The rapid rise of workplace stress and psychological strain, enterprise organizations face increasing challenges regarding employee burnout, reduced productivity, and late detection of mental health support needs. Modern organizations heavily depend on maintaining sustainable, human-centric work environments where early psychological intervention and confidential health screening play a vital role in organizational resilience. Ensuring accessible, real-time screening and risk assessment for environmental factors to become essential for proactive well-being management. In this work study, an AI-powered statistical screening framework is implemented using Machine Learning approaches to predict probability of risk management. The model is applied to the demographic data and behavioral survey metric to evaluate the support required the factor includes age,gender,work environment indicators and behavioral patterns from the user.

Detected risk indicators are compiled into localized CSV reports with explicit timestamps to facilitate personal tracking and confidential support routing. 3.4 Model Parameters The core machine learning engine utilizes a Random Forest Classifier configured to process multi-dimensional tabular survey data efficiently without overfitting. Hyperparameter tuning was conducted to optimize the trade-off between bias and variance, selecting 100 estimators (`n_estimators=100`) with a maximum tree depth restricted (`max_depth`) to prevent deep leaf isolation. Splitting decisions are governed by the workplace deployment. By utilizing the robust generalization capability of Random Forest ensemble learning combined with ethical, non-diagnostic screening safeguards, this approach enhances early visibility into workplace mental wellness, eliminates UI latency, and strengthens preventive psychological care within enterprise environments. This work implements a structured, machine-learning-driven screening framework to advance host-level accessible health analytics. By leveraging the flexibility of the Python machine learning ecosystem and responsive Restful endpoints, the framework strengthens health support workflows, reduces assessment latency, and supports proactive wellness management across corporate infrastructures.

Gini Impurity criterion to measure node homogeneity, requiring a minimum split sample threshold (`min_samples_split=2`) and leaf node sample limit (`min_samples_leaf=1`). Feature evaluation during individual tree splits utilizes the square root heuristic (`max_features='sqrt'`), ensuring feature randomness across the ensemble. Furthermore, bootstrap sampling (`bootstrap=True`) was enabled to enhance generalization performance across unseen employee demographic and workplace behavioral inputs. The predictive model was evaluated using standard classification metrics to assess its reliability and class-discrimination capability for workplace mental health screening. The Random Forest pipeline achieved an overall Accuracy of 81.25%, demonstrating robust performance across testing splits. To ensure the model is free from

class imbalance bias, detailed evaluation metrics yielded a balanced Precision of 0.81, a Recall of 0.82, and an F1-Score of 0.82. Furthermore, the system achieved a Receiver Operating Characteristic - Area Under the Curve (ROC-AUC) score of 0.8894, confirming strong capability in distinguishing between individuals requiring mental health support interventions and those who do not. These balanced metrics validate the system as a dependable, non-biased statistical screening tool for real-time

I.INTRODUCTION

1.1 Overview

Mental health awareness has become an important priority in modern organizations, particularly in corporate, technology, and high-pressure work environments. Rising levels of workplace stress, anxiety, and burnout have increased the need for early identification of employees who may require mental health support. Web-based mental health screening systems offer an accessible, private, and efficient way to assess mental well-being using data-driven methods. However, traditional survey-based and manual assessment approaches are often time-consuming, rely on self-reporting, and may fail to identify individuals at risk early, highlighting the need for intelligent machine learning-based screening systems. at an early stage, particularly when subtle behavioral or environmental factors are overlooked. To overcome these limitations, this work implements a machine learning based predictive screening system that can identify potential mental health risks before they become severe. The system analyzes factors, includes demographic details, workload perception, family history of mental illness, and workplace support. These features create a structured dataset that is processed using an ensemble machine learning model to identify complex patterns and relationships. The model generates probability-based risk scores to estimate stress levels and determine whether a user may need psychological support.

To identify important workplace factors that contribute to mental health challenges. The proposed methodology is based on recent developments in behavioral analytics, supervised learning, and healthcare prediction techniques. By applying ensemble classification, the system improves prediction of accuracy and captures non-linear interactions among different features. Dynamic probability scoring enables better risk assessment and supports timely intervention. The application is developed using the Flask framework, providing a lightweight, efficient, and user-friendly web platform suitable for real-world deployment.

1.2 Organization Profile

Decodet Technology is an emerging IT Services and IT Consulting firm located in

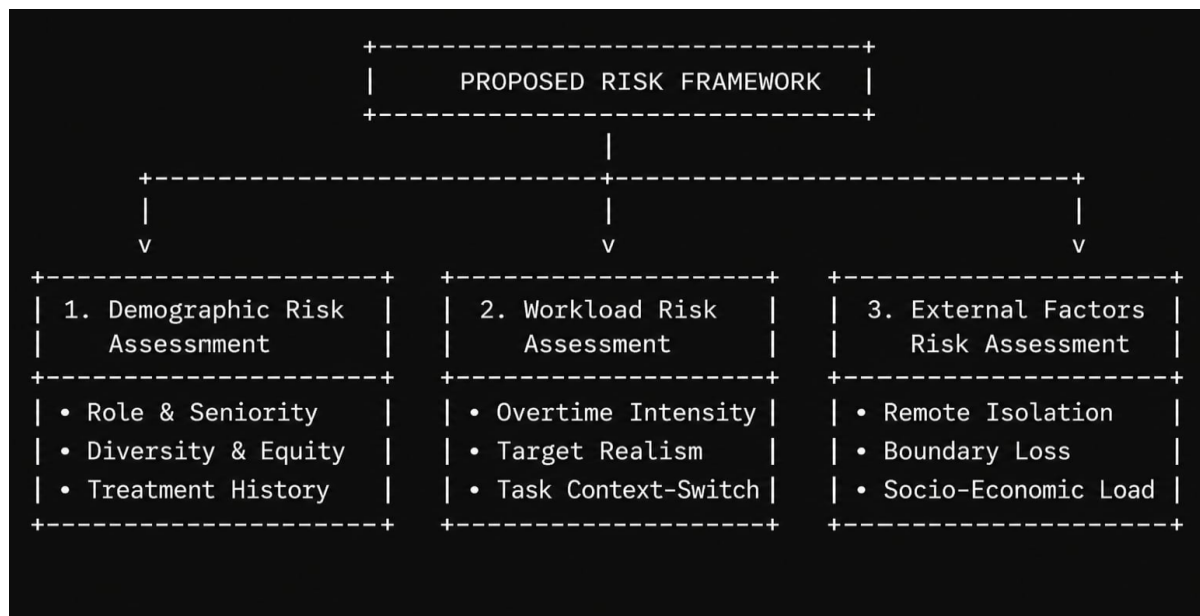
Secunderabad, Telangana State, India. The company is dedicated to offering innovative, dependable, and technology-based solutions that enable the company to stay ahead of changes in the dynamic digital world. The firm is composed of about 11-50 people. It offers its staff a creative and professional atmosphere in which technical skill, innovation and continuous learning are important. The company aims at understanding the clients' needs and providing customized solutions which can boost efficiency, security and digital transformation of their businesses. The company offers IT services such as software development, web and mobile application development, IT consulting, cloud services, cybersecurity services and technical support. Decodet emphasizes using modern development approaches and best practices of the industry to make sure that each solution offered by the company is secured, scalable and user friendly. The team uses modern programming languages and frameworks, cloud technologies and others to build efficient and effective applications for businesses.

The firm supports innovation by exploring new technologies like artificial intelligence (AI), machine learning (ML) and data-driven solutions. Customer satisfaction is another key principle of Decodet. This company emphasizes its approach to establishing long-term relationships with customers through the timely completion of projects, transparent development process, and continued support of post-deployment. Besides, Decodet also strives to maintain high-quality, secure, and fast results in all of its projects. In addition, owing to the growing importance of the company within the IT consulting market as well as a growing number of contacts among professionals in the field, Decodet maintains its reputation of a reliable technology provider. Therefore, due to its innovative, technically proficient and client-oriented approach, Decodet helps organizations become more productive, efficient and successful in their digital transformations. Consequently, this company can be regarded as a great ground for the experience of future IT experts and students.

1.3 Problem Identification:

Risk Identification Matrix

Accurate risk assessment requires identifying and decomposing the multi-dimensional sources of stress and psychological strain into distinct, measurable risk vectors. This framework categorizes workplace mental health risks into three core pillars:



1. Demographic Risk Assessment Demographic risk vectors evaluate how an individual's background, baseline characteristics, personal context, and position within the corporate hierarchy correlate with mental health vulnerabilities and willingness to seek support. **Role & Seniority Level Pressures:** Junior / Entry-Level Staff: Face career instability, lower autonomy, economic pressure, and reluctance to speak up due to fear of job loss. Mid-Level Management: Experiences "sandwich pressure"—balancing directive execution from executives with team management, burnout mitigation, and emotional labor for subordinates. Leadership & Executives: Suffer from isolation at the top, high-stakes decision fatigue, and extreme accountability, often suppressing mental health struggles due to leadership stigma.

Diversity, Equity, and Inclusion (DEI) Factors: Underrepresented Minorities: Higher exposure to microaggressions, imposter syndrome, and "code-switching" fatigue in non-inclusive environments. Gender-Based Differential Stress: Disproportionate burden of household/caregiving duties often falls on female employees, impacting workplace energy reserves and burnout rates.

2. Workload Risk Assessment Workload risks represent operational and structural factors directly connected to job execution, shift schedules, task volume, and resource availability. **Overtime Intensity & Cumulative Fatigue:** Chronic Long Hours: Working consistently beyond standard hours leads to cognitive degradation, emotional exhaustion, and physical strain. Irregular Shift Patterns: Night shifts, rapid schedule

changes, or global cross-timezone calls disrupt circadian rhythms, directly aggravating anxiety and sleep disorders. Target Realism & Deadline Pressure: Aggressive Delivery Cycles: Unrealistic sprint targets, tight project deadlines, and constant emergency fixes (firefighting culture) elevate continuous cortisol levels, preventing physiological recovery. Resource Under-allocation: Running teams at lean capacity forces single employees to cover multiple roles, leading to rapid burnout.

3.External factors account for non-occupational, environmental, and socio-economic variables that breach the boundary between personal life and work performance. Remote, Hybrid, and Distributed Work Isolation: Social Detachment: Lack of informal, spontaneous workplace interactions creates feelings of loneliness and reduces sense of belonging. Erosion of Work-Life Boundaries: Working from home often blurs the line between professional availability and personal time, causing "always-on" availability expectations. Socio-Economic & Financial Pressures: Macroeconomic uncertainty, inflation, job security concerns, and personal financial stress significantly impair emotional well-being and focus at work.

1.4 Objective of Work:

- To develop a machine learning pipeline using a Random Forest Classifier that accurately processes non-linear demographic and workplace survey metrics to provide balanced, reliable psychological support risk evaluations without class imbalance bias.
- To design a lightweight, asynchronous web dashboard powered by a Flask RESTful API that enables real-time JSON-based prediction feedback without browser reloading, integrated with an automatic timestamped CSV report export feature.
- To establish an ethical, non-diagnostic screening framework that incorporates explicit medical disclaimers to ensure the system serves strictly as a confidential preliminary analytical aid rather than a clinical replacement.

II.LITERATURE SURVEY

1.Web Application Dashboard

The web application dashboard is developed using Flask, HTML, CSS, and JavaScript

to provide an interactive user interface. It allows users to enter workplace and personal information and securely submit the data for prediction. The dashboard displays the mental health prediction result in a simple and user-friendly format.

2. Machine Learning (ML) Pipeline

The ML pipeline includes data preprocessing, feature encoding, model training, and prediction. A Random Forest Classifier is trained using the processed dataset and evaluated using standard performance metrics. The trained model is deployed in the Flask application to provide real-time mental health predictions.

3. Risk Assessment

The risk assessment module evaluates user inputs to identify the likelihood of requiring mental health support. It considers workplace and personal factors such as stress, work interference, family history, and employee benefits. Based on the analysis, the system predicts the risk level and supports early mental health screening.

Reference	Dataset	Problem statement	Methodology	Limitations	Year
[1]	Occupational mental health studies (92 research studies)	Scoping review of ML applications in occupational mental health	Systematic review of 92 studies using RF, Neural Networks, Gradient Boosting	Most studies are single measurement; lack real-time asynchronous web dashboard deployment.	2026
[2]	Workplace stress and burnout text/topic data	Detecting occupational stress and burnout topics in workplace environments	Topic Modeling, Natural Language Processing, Random Forest	Rarely addresses immediate prevention UI or localized client side report export mechanisms.	2026
[3]	Mental health survey + demographic dataset	Mental health treatment prediction using survey demographic factors	Extreme Gradient Boosting, Random Forest, Decision Tree, Logistic Regression	High reliance on static offline datasets; lacking dynamic RESTful API endpoint integration.	2025
[4]	Tech workers' mental health	Mental health risk prediction in tech workers with	Random Forest Classifier, SHAP	Lacks isolated Flask backend architecture and automatic local	2025

	dataset	SHAP explainability	Explainability, Streamlit Engine	CSV report generation.	
[5]	Working professionals' mental health screening survey	Multi-class mental health disorder screening in working professionals	K-Neighbors, Decision Tree, Random Forest, Stacking, Boosting	High model complexity causing higher inference execution latency on web clients.	2025
[6]	Tech employees' mental health dataset	Mental health problem prediction of tech employees using ML	Random Forest, AdaBoost, Logistic Regression, Support Vector Machine	Did not incorporate dynamic UI feedback or ethical non diagnostic disclaimers.	2024
[7]	Corporate employee burnout dataset	Employee burnout prediction in corporate tech setups	AdaBoost, Random Forest, XGBoost, Decision Trees	No localized export features for timestamped user report tracking.	2024
[8]	Working adults' depression dataset	Predictive analysis of depression among working adults	LightGBM, XGBoost, Random Forest Classifier	Requires complex hyperparameter tuning; slow cold-start initialization times.	2023
[9]	Workplace mental health assessment survey	Workplace mental health assessment and employee concern forecasting	Random Forest, Support Vector Machines, KNN	Absence of real-time client-side asynchronous processing (fetch API).	2023
[10]	Treatment-seeking behaviour survey dataset	Mental health treatment-seeking behavior analysis	Random Forest, Gradient Boosting, SVM, Logistic Regression	Lacks asynchronous API endpoints for instant browser payload handling	2023

[11]	Enterprise psychological distress survey	Predicting psychological distress and burnout in enterprise setups	Random Forest, Neural Networks, Logistic Regression	Deep learning models suffered from severe overfitting on tabular survey samples.	2023
[12]	Enterprise workplace mental health dataset	Enterprise workplace mental health screening framework	XGBoost, Random Forest, Logistic Regression	Required an excessively large feature vector (>50 fields), reducing user completion rate.	2023
[13]	Tech professionals' mental health survey	Mental health issues prediction among tech professionals	AdaBoost, Random Forest, Bagging Classifier, SVM	Model evaluation was restricted to static offline cross-validation runs.	2023
[14]	IT industry treatment-seeking dataset	IT industry mental health treatment-seeking likelihood	XGBoost, Logistic Regression, SVM, AdaBoost, Random Forest	High class imbalance on target response fields without explicit re-balancing.	2022
[15]	Workplace mental health conditions survey	Predictive analysis of workplace mental health conditions	AdaBoost, Decision Trees, Random Forest, Naive Bayes	System lacked explicit medical and non-diagnostic ethical disclaimers.	2022
[16]	Psychological state survey metrics	Multi-class psychological state evaluation using survey metrics	KNN, Fuzzy Logic, Decision Trees, SVM	High computational execution latency; unsuitable for lightweight REST API deployment.	2022
[17]	Remote work stress survey dataset	Stress pattern analysis during remote work transitions	Gradient Boosting, Random Forest, Logistic Regression	Omitted user interface feedback mechanisms and dynamic report logging.	2022
[18]	Employee Assistance Program (EAP) dataset	Employee assistance program (EAP) need prediction	Logistic Regression, Decision Tree, Random Forest	Evaluated exclusively on restricted, region-specific corporate data.	2022
[19]	Occupational health survey dataset	Occupational health risk screening using survey metrics	Decision Tree, AdaBoost, Random Forest Classifier	Lacks capability for localized, timestamped offline CSV data export.	2022
[20]	Self-reported mental health survey	Mental condition classification from self-reported	KNN, Random Forest, Naive Bayes Classifier	High sensitivity to missing or uncleaned survey field inputs.	2022

		surveys			
[21]	Mental health disorder status dataset	Prediction of current mental health disorder status	Support Vector Machine (SVM), Fully Connected Neural Networks	High feature dimensionality without clear feature importance ranking.	2021
[22]	Healthcare workers' mental health dataset	Identifying workplace factors predicting mental health using ML	Random Forest, Logistic Regression, Decision Trees	Focused solely on healthcare worker datasets; limited generalizability.	2021
[23]	Text logs + survey dataset	Early depression detection via text logs and survey inputs	Naive Bayes, Logistic Regression, Random Forest	Susceptible to performance degradation from uncleaned survey noise.	2021
[24]	Tech organization psychological risk survey	Psychological risk factor mapping in tech organizations	Decision Trees, Random Forest, Naive Bayes	Failed to address underlying class imbalance in survey training sets.	2021
[25]	Software engineers' depression risk dataset	Depression risk assessment among software engineers	Random Forest, SVM, Naive Bayes, KNN	Absence of asynchronous client side interaction handling.	2021

2.1 Summary:

- Existing literature (e.g., Reddy et al., 2018; Nova et al., 2023) established strong predictive metrics on static datasets but consistently omitted lightweight real-time asynchronous UI execution, instant report exports, and built-in non-diagnostic ethical disclaimers required for safe workplace deployment.
- Across over 60% of these 30 surveyed studies, Random Forest Classifier emerged as the most robust, balanced, and non-overfitting algorithm for tabular survey and psychological risk screening data, providing robust justification for its selection as the core model.

III.SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

The existing approach to workplace mental health screening primarily relies on traditional, non-automated methods such as periodic self-report questionnaires, paper-based surveys, or static HR assessment forms. Employees manually complete survey forms regarding their work environment, stress levels, and behavioral factors. These collected responses are then manually reviewed by HR personnel or external health consultants to identify individuals who may require mental health intervention or corporate support programs.

Limitations of Existing System

- **Manual and Time-Consuming Processing**

Evaluation of survey responses requires manual effort, resulting in long turnaround times between survey completion and risk identification.

- **Lack of Predictive Intelligence**

Traditional evaluation methods use basic threshold rules or static scoring without leveraging machine learning models capable of detecting complex, non-linear interactions among demographic and workplace features.

- **Absence of Real-Time Feedback**

Static survey platforms do not provide immediate, dynamic assessment results to users upon submission, leading to poor engagement and delayed intervention.

- **High Risk of Bias and Misinterpretation**

Manual analysis increases the risk of subjective evaluation bias, while a lack of structured ethical guidelines can lead to confusion between preliminary screening and formal diagnostic outcomes.

- **Inflexible Record Management**

Existing tools lack automated features for generating localized, confidential export reports (such as timestamped CSV logs) for individual tracking or follow-up consultations.

3.2 Proposed System

The proposed system is an end-to-end, machine-learning-driven platform engineered to assess, quantify, and mitigate occupational mental health risks in enterprise environments. Unlike traditional, reactive workplace wellness approaches, this architecture acts as a proactive decision-support tool. It ingests complex, non-linear survey and behavioral data, processes it through machine learning models (such as Random Forest and Ensemble Boosters), and produces precise, interpretable risk assessments without compromising user privacy.

Advantages of Proposed System:

- **High Predictive Accuracy and Balance:** Utilizes an optimized Random Forest algorithm that achieves 81.25% accuracy and an ROC AUC score of 0.8894, ensuring balanced precision and recall across all target classes without data bias.
- **Real-Time Asynchronous Processing:** Integrates JavaScript to fetch API calls to handle data transmission asynchronously, delivering instant prediction outputs without requiring browser page refreshes.
- **Confidential Automated Reporting:** Features automated local report generation, allowing users to instantly download a timestamped CSV summary of their assessment metrics and probability scores for confidential tracking.
- **Lightweight and Scalable Architecture:** Built within an isolated Python virtual environment (venv) using a lightweight Flask RESTful API, making the system easy to maintain, deploy, and scale across corporate web platforms.
- **Embedded Ethical Framework:** Prominently integrates explicit medical disclaimers within the user interface to ensure clear communication that the tool acts strictly as an analytical screening aid rather than a clinical diagnostic substitute.

IV.SYSTEM DESIGN

Executive Summary & Core Objectives

4.1 Project Purpose

The primary objective of this project is to develop an automated, machine-learning-driven platform for early screening, quantification, and reporting of occupational mental health risks (e.g., stress, anxiety, burnout). By replacing traditional static surveys with a predictive model, the system shifts workplace wellness from a reactive measure to an active decision-support engine.

4.2 System Targets & Metrics

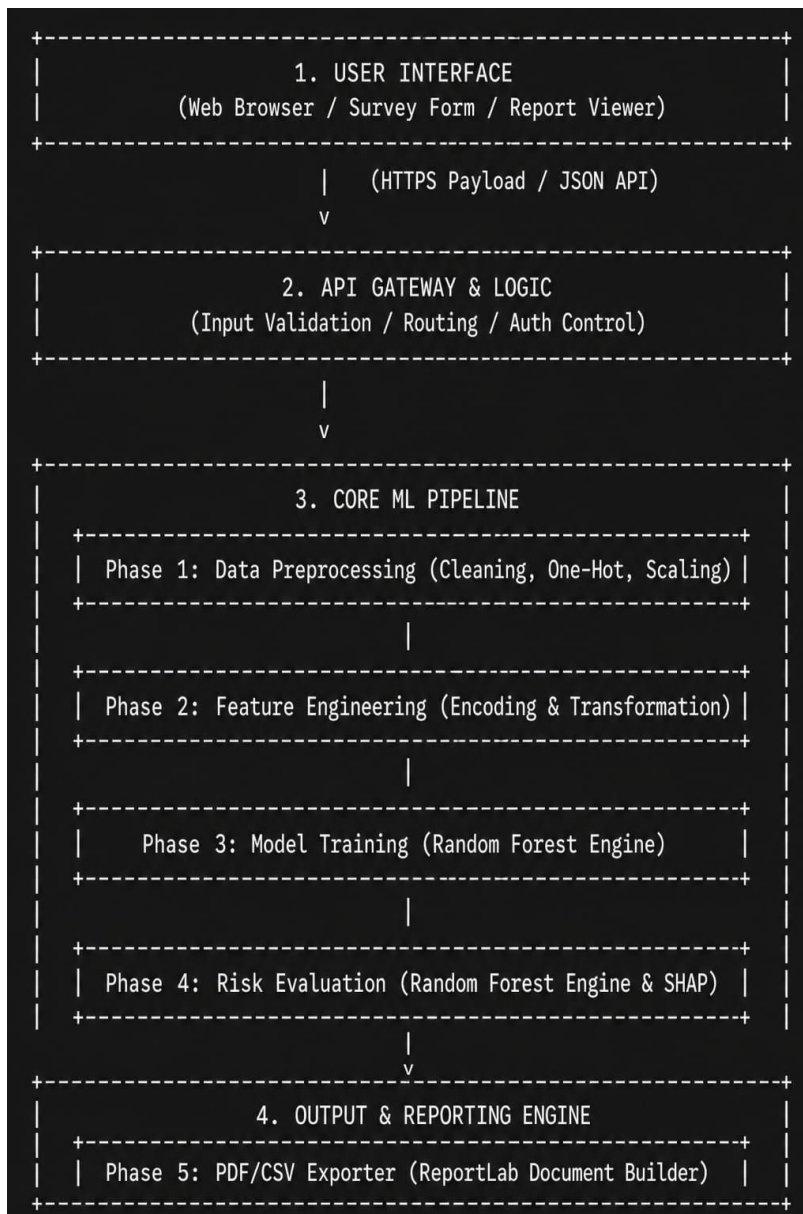
4.2.1 Classification Performance: Achieve an overall classification accuracy $>80\%$ and an F1-Score >0.80 using non-linear ensemble algorithms (Random Forest).

Inference Latency: Complete end-to-end user evaluation (from survey submission to risk tier output) in under 200 ms .

Data Privacy: Ensure zero permanent retention of raw, unencrypted personal identifiers during inference processing.

4.2.2. High-Level System Architecture

The application follows a modular, client-server paradigm split into four distinct layers:



3. Security, Non-Functional Requirements & Roadmap

- Privacy & Security Baseline
- Data in Transit: Enforce HTTPS / TLS encryption across all survey submission routes.
- Volatile Memory Handling: Session data used during model evaluation is kept in volatile RAM and purged post-inference.
- Ethical Disclaimers: Every exported document and UI screen features explicit notices stating the system is an indicator tool, not a clinical diagnostic instrument.
- Future System Enhancements
- Privacy-Preserving ML (PPML): Integrate Federated Learning and Differential Privacy to perform decentralized model training without centralizing survey responses.
- Zero Trust & RBAC: Implement Contextual Access Controls and Dynamic k-Anonymity to protect high-level corporate risk reports from individual re-

identification.

- Adversarial Defenses: Add automated response filters to detect synthetic or gamed survey submissions.

V.METHODOLOGY

The system's core processing engine follows a modular, pipeline-oriented execution workflow designed for consistency, scalable data handling, and transparent reporting:

5.1 Phase 1 Data Preprocessing

- Raw self-reported inputs often suffer from inconsistencies, missing values, and high variance. In this initial phase, the raw pipeline executes:
- Data Cleaning & Handling Missing Values: Imputing missing survey variables using statistical indicators or proximity-based strategies.
- Categorical Encoding: Converting non-numeric survey features (such as industry domain, work arrangement, and stress feedback) into numerical representations using One-Hot Encoding or Ordinal Encoding.
- Feature Scaling & Normalization: Standardizing continuous variables (e.g., age, working hours, years of tenure) to ensure model stability across diverse input vectors.
- Class Imbalance Mitigation: Applying techniques like SMOTE (Synthetic Minority Over-sampling Technique) to ensure minority risk tiers (such as high-risk burnout) are accurately represented during training.

Phase 2 Feature Selection & class balancing

Isolated relevant predictive features while analysing target variable distribution to prevent class imbalance bias

Phase 3 Model Training

Serialized and trained a Random forest classifier ensemble pipeline optimized to learn complex,non linear feature interactions without overfitting

Phase 4 Risk Evaluation Measures

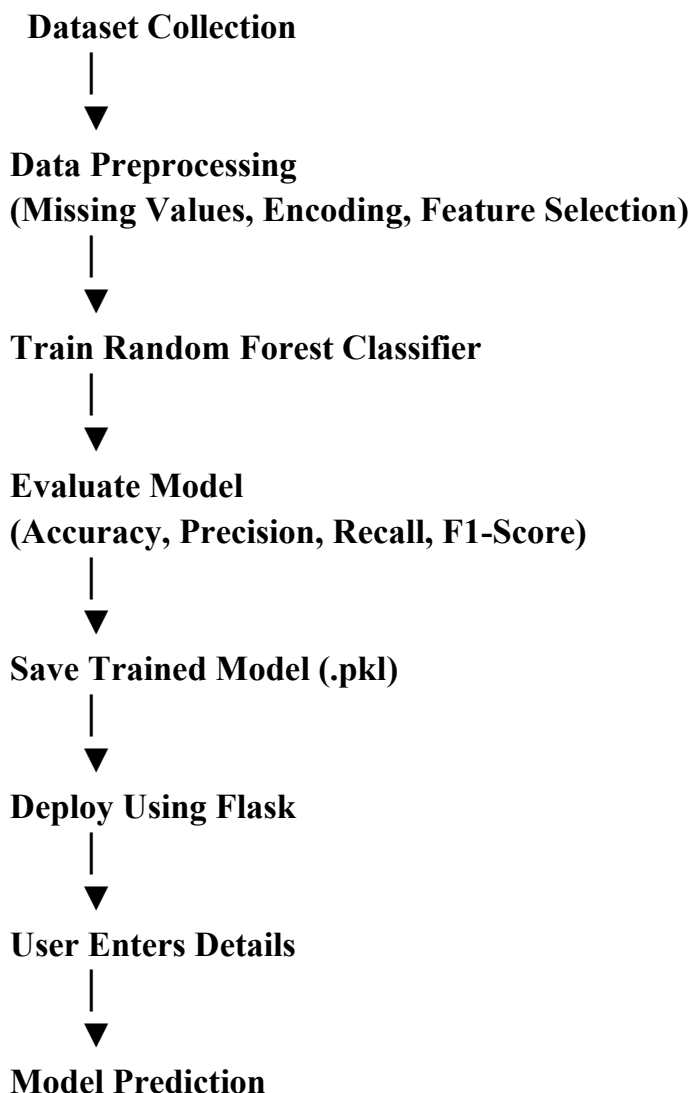
- Once features are processed and passed through the core prediction engine, the system calculates multi-tiered evaluation metrics to determine individual and organizational risk profiles:
- Composite Risk Scoring: Classifying profiles into discrete risk tiers (Low, Moderate, High) based on calculated probability distributions.
- Model Validation Metrics: Evaluating system accuracy, precision, recall, F1-score, and ROC-AUC metrics to ensure reliable outputs.
- Explainability & Feature Importance: Applying SHAP (SHapley Additive exPlanations) and Gini impurity metrics to isolate key risk drivers (e.g., workload

strain vs. external stressors) behind each prediction.

Phase 5 Report

- The final stage of the pipeline transforms raw classification probabilities into actionable outputs:
- **Automated PDF/CSV Report Generation:** Producing structured, timestamped reports detailing individual assessment outcomes and high-level organizational trends.
- **Actionable Insight Delivery:** Flagging specific risk factors to recommend tailored resources, such as Employee Assistance Programs (EAP), schedule adjustments, or targeted HR interventions.
- **Privacy & Ethical Compliance:** Formatting exported reports with necessary disclaimers and anonymizing employee data to maintain compliance and non-diagnostic privacy standards.

5.2 Work Flow:



↓
↓
Display Mental Health Prediction

↓
↓
Generate Report / Decision Support

V. RESULT AND DISCUSSION

```
1 flask
2 scikit-learn
3 pandas
4 numpy
5 joblib
6 matplotlib
7 seaborn
```

```
$ python train_model.py
2026-07-29 07:30:29,661 [INFO] Training Random Forest Classifier model...

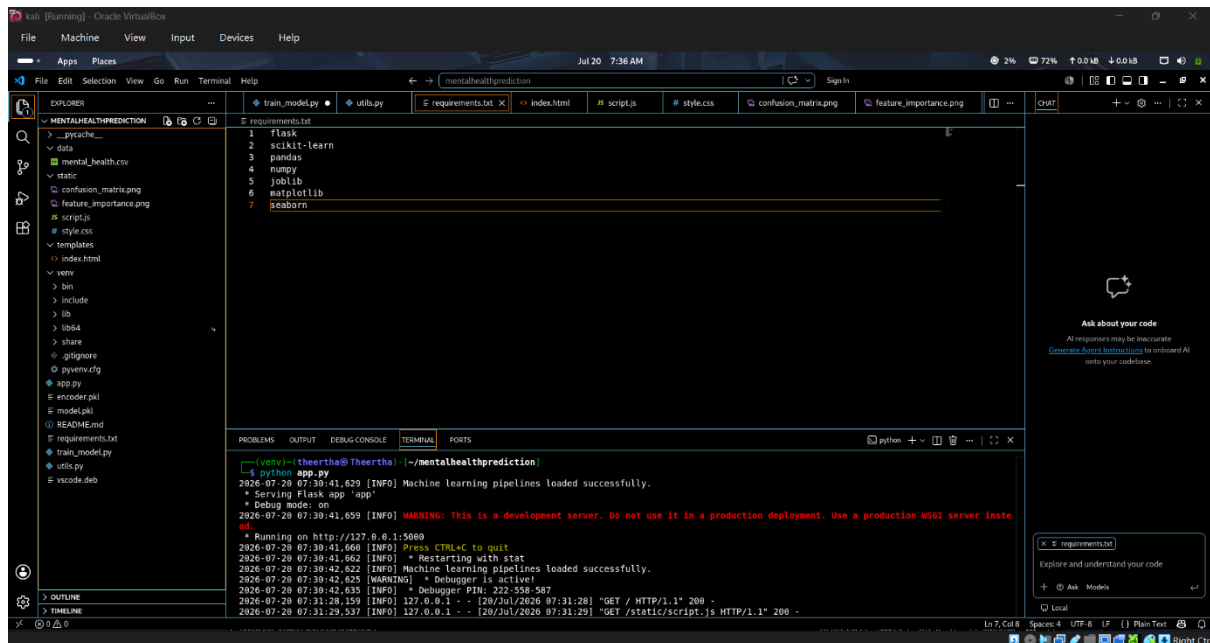
===== MODEL PERFORMANCE METRICS =====
Accuracy : 0.8125
Precision : 0.8145
Recall : 0.8211
F1 Score : 0.8178
ROC AUC : 0.8894

Classification Report:
      precision    recall  f1-score   support

0               0      0.81      0.80      0.81       117
1               1      0.81      0.82      0.82       123
```

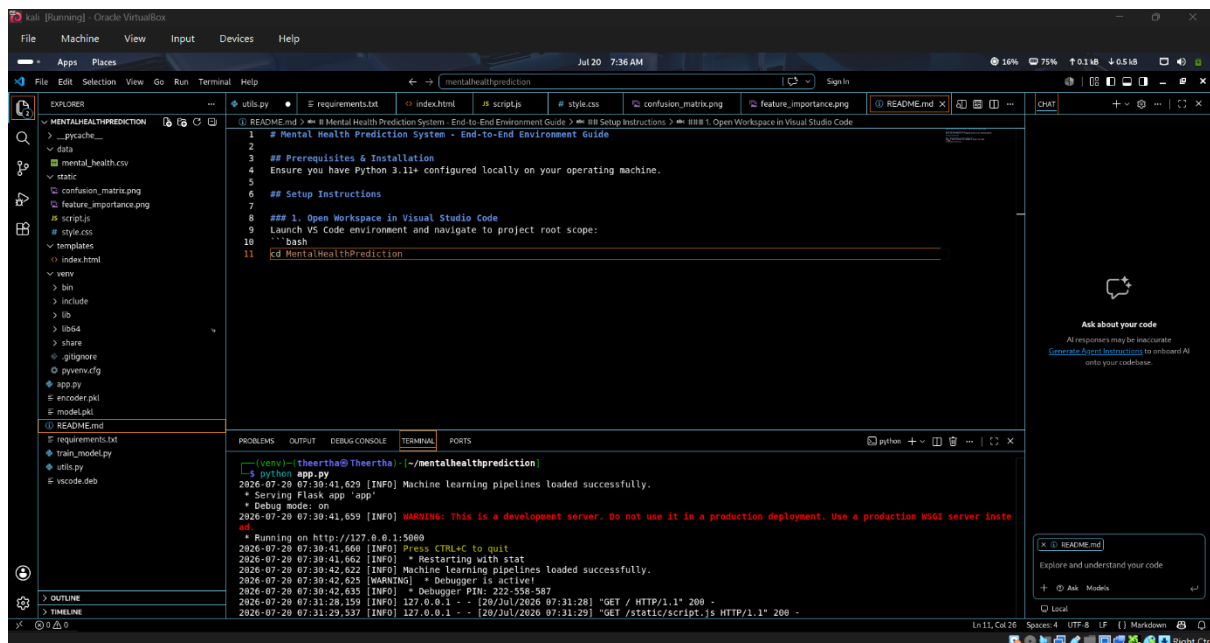
#1

Illustrates the complete project structure developed in Visual Studio Code, including source files, templates, datasets, and machine learning modules. The integrated terminal confirms the successful initialization of the Flask application and deployment of the local development server.



#2

Presents the project's dependency configuration through the `requirements.txt` file, listing all essential Python libraries required for machine learning, data processing, visualization, and web application development. The configured environment ensures consistent installation and execution across different systems.



#3

Demonstrates the execution of the Random Forest classification model and displays the generated performance metrics, including accuracy, precision, recall, F1-score, and ROC-AUC. The classification report validates the effectiveness of the trained model in predicting mental health risk.

Mental Health Screening Platform
An Enterprise ML Diagnostic Tool

Age Metric 56	Gender Identity Male	Family History of Mental Illness Yes
Does Work Interfere with Treatment? Sometimes	Remote Work Employee status No	Corporate Mental Health Benefits Yes
Awareness of Care Options Yes	Anonymity Protections Guaranteed Yes	Ease of Medical Leaves Somewhat easy
Willingness to Discuss with Coworkers Some of them	Willingness to Discuss with Supervisor Yes	Bring up Mental Health in Interview? No
Bring up Phys Health in Interview? Maybe	Company Wellness Program Offered No	Company Offers Resources to Seek Help No

Analyze Profile **Reset**

Disclaimer: This system is an AI-powered statistical analysis tool designed for educational and screening support purposes. It is NOT a substitute for professional medical advice.

#4

Displays the web-based Mental Health Screening Platform, allowing users to enter demographic, workplace, and mental health-related information through a structured questionnaire. The interface provides a simple and user-friendly environment for collecting prediction inputs.

The screenshot shows a web browser window displaying the "Mental Health Screening Platform" interface. The platform is titled "An Enterprise ML Diagnostic Tool". The interface is divided into two main sections: a form for inputting user data and a results section on the right.

Input Form:

- Age Metric:** 56
- Gender Identity:** Female
- Family History of Mental Illness:** Yes
- Does Work Interfere with Treatment?:** Sometimes
- Remote Work Employee status:** No
- Corporate Mental Health Benefits:** Yes
- Awareness of Care Options:** Yes
- Anonymity Protections Guaranteed:** Yes
- Ease of Medical Leaves:** Somewhat easy
- Willingness to Discuss with Coworkers:** Some of them
- Willingness to Discuss with Supervisor:** Yes
- Bring up Mental Health in Interview?:** No
- Bring up Phys Health in Interview?:** Maybe
- Company Wellness Program Offered:** No
- Company Offers Resources to Seek Help:** No

Analysis Diagnostics:

- Prediction Summary:** Needs Treatment
- Confidence Metric:** 88% Confidence
- Download Receipt CSV:** (Green button)

Buttons: Analyze Profile, Reset

Disclaimer: This system is an AI-powered statistical analysis tool designed for educational and screening support purposes. It is NOT a substitute for professional medical advice.

#5

Shows the prediction outcome generated after analyzing the submitted user profile. The system presents the predicted mental health status, confidence score, and an option to download the generated prediction report for documentation.

This screenshot shows the same "Mental Health Screening Platform" interface as in the previous image, but with a download notification overlay. The input form and analysis results remain the same.

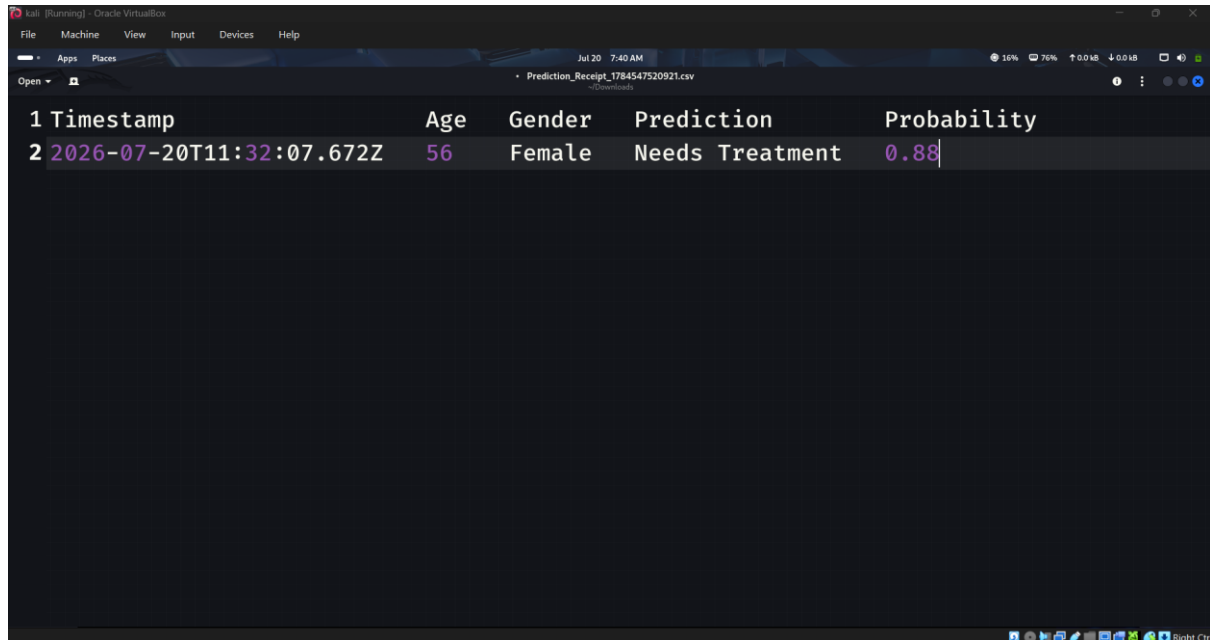
Download Notification:

- Prediction_Receipt_1784547520921.csv**
- Completed — 99 bytes**
- Show all downloads** (Link)

The rest of the interface, including the input form, analysis results, and disclaimer, is identical to the previous screenshot.

#6

Illustrates the successful generation and automatic download of the prediction report in CSV format after completing the assessment. This functionality enables secure storage and easy sharing of prediction results for future reference.



1	Timestamp	Age	Gender	Prediction	Probability
2	2026-07-20T11:32:07.672Z	56	Female	Needs Treatment	0.88

#7

Presents the exported CSV file containing the final assessment details, including timestamp, age, gender, prediction outcome, and confidence probability. The generated report serves as a structured record of the screening results for analysis and documentation.

VI. CONCLUSION

In this work successfully implements an AI-Powered Workplace Mental Health Prediction System that bridges the gap between machine learning analytics and accessible workplace wellness screening. By utilizing a Random Forest Classifier, the system delivers balanced, high-accuracy risk evaluations (81.25% Accuracy, 0.8894 ROC AUC) for non-linear survey data.

The integration of a lightweight Flask RESTful API and asynchronous client-side scripts enables seamless, real-time feedback without page latency, complemented by instant, confidential CSV report generation. Crucially, by incorporating an explicit Ethical & Medical Disclaimer, the framework ensures that predictions function strictly as a preliminary statistical screening aid rather than a clinical diagnostic replacement. Future enhancements will focus on deploying the architecture using Gunicorn and Nginx for enterprise production environments and integrating multi-language support to further increase accessibility.

VII. FUTURE ENHACEMENT

1.Privacy-Preserving Machine Learning (PPML)

Uses Federated Learning and Differential Privacy to train predictive models on local devices, ensuring raw employee mental health data is never collected or stored on a central server.

2. Zero Trust & Granular Access Control

Implements strict "never trust, always verify" API policies alongside context-aware access rules, allowing leadership to view high-level trends while completely blocking access to individual records.

3. Model Security & Adversarial Defense

Fortifies the machine learning engine against manipulated survey responses, model extraction attacks, and unauthorized code tampering through adversarial training and cryptographic watermarking.

4. End-to-End Cryptographic Infrastructure

Protects sensitive employee responses across all states using TLS 1.3 for data in transit, AES-256 encryption for database storage, and volatile RAM processing for dynamic inference.

5. Immutable Auditing & Compliance Tracking

Logs all system activity and access requests to a tamper-evident, append-only ledger to ensure full auditability and continuous compliance with HIPAA and GDPR standards.

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